

Fitness Management System Project Report

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Project Overview

The Fitness Management System (branded as Liftbite) is a simple, console-based application developed using Python. The project is designed to show basic Python programming concepts through a practical fitness evaluation application.

This project doesn't use a GUI, database, internet connection, or any external service. All interactive evaluations and profile parameters are processed entirely locally.

Features

The system provides the following features:

User Profile Management

- Create a new user fitness profile
- Validate input for name, age, weight, and height
- Keep information organized dynamically during execution

Fitness & Health Assessment

- Dynamic Body Mass Index (BMI) computational logic
- Automated threshold classification mapping
- Conditional age-restriction checks (e.g. warnings for individuals under 18)

Goal-Oriented Action Routine

- Four distinct customized fitness goals: Gain, Lose, Maintain, Abs
- Specific equipment mapping arrays
- Targeted activity recommendation schedules

Medical Profiling & Precautionary Triggers

- Active logging of medical histories
- Automatic injection of custom clinical warnings and boundaries

Session Summary & Receipt Log

- Dynamic receipt output summarizing the current profile session
- Local date and time stamps using standard Python engines

Technologies Used

- Python 3
- Python standard library modules (calendar, datetime, random)
- Core loop handlers, numeric filtering arrays, text formatting scripts

No external Python packages are required.

Project Structure

Liftbite_Fitness_System/

- main.py (Interactive entry script containing user profile loops & evaluation logic)
- README.md (Project manual describing run procedures and hardware limits)

Module Description

1. main.py

This is the main program file. It welcomes the user, triggers input filters, builds the profile structure, determines BMI parameters, assigns appropriate fitness routines, prints an execution receipt, and manages structural session looping workflows.

Requirements

To run this project, you need:

- Python 3.x
- Any Python editor or console engine (VS Code, PyCharm, IDLE, Thonny)

No internet connection is required.

How to Run

Step 1: Create a folder named 'Liftbite_Fitness_System'

Step 2: Place 'main.py' inside the folder

Step 3: Open terminal within that path and run: `python main.py`

Program Run Execution Logs (Output Mock Screen)

```
=====
                        Welcome to Liftbite
A simple starting point for your fitness routine
Let's get started.
=====

Let's set up your profile.
What should we call you? Kushal Farkya
How old are you? 19
Your weight in kilograms: 70
Your height in centimetres: 175

Choose your main goal:
1. Gain weight and build strength
2. Lose weight and improve fitness
3. Maintain my current weight
4. Work on my abs
Enter 1, 2, 3, or 4: 2
Do you have any medical history? (yes/no): no

Your Liftbite profile
-----
Name: Kushal Farkya
Age: 19
Weight: 70.0 kg
Height: 175.0 cm
Goal: lose
Medical history: None reported
Profile created: Wednesday, 2026-09-30

BMI: 22.9
General category: Within the usual adult range

A few ideas to get started
-----
BMI for your selected goal: 22.9
1. Keep meals varied, with vegetables, fruit, whole grains, and protein.
2. Mix activities you enjoy, such as brisk walks and light strength work.
3. Build a routine you can keep up, and make room for rest.

Equipment and exercises for your goal
Equipment: walking shoes, a skipping rope, and an exercise mat.
Exercises: brisk walking, cycling, step-ups, and bodyweight exercises.

A quick note
BMI is a rough screening measure, not a diagnosis or a full picture of health.

Thanks for stopping by Liftbite, Kushal Farkya!

Receipt
Name: Kushal Farkya
Date: 2026-09-30
Time: 03:15:22
Weight: 70.0 kg
Height: 175.0 cm
Goal: lose
Medical history: None reported
BMI: 22.9
Category: Within the usual adult range
Thank you for using Liftbite

Do you want another session? (yes/no): no
```

Goodbye !

=====

Python Concepts Demonstrated

- Variables and native data types
- Multi-tier conditional branches (if, elif, else)
- Interactive loop routines (while loops)
- Custom computational function structures
- Basic validation rules and numeric type formatting
- Date and time handling

Project Objectives

- Build structural multi-layered programs in Python
- Establish custom numerical validation controls
- Formulate practical application tools mapping clinical parameters

Limitations

This is an educational project and isn't intended for real banking or real clinical use. The project does not utilize external database backends, online web hosting channels, or cryptographic security layers.

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.